

Design and Verification Report: Vending Machine Controller

Digital Logic Design Documentation

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1 Introduction

This report describes the design flow, internal working, and verification process for a VHDL-based vending machine controller. The system accepts 5-cent and 10-cent coins and dispenses water when a total of 15 cents is reached, providing change if necessary. It also features a safety timeout mechanism.

2 Design Flow

The design follows a standard Digital System Design Flow:

1. **Requirement Analysis:** Identification of inputs (coins, clock, reset) and outputs (water, change).
2. **RTL Modeling:** Writing the VHDL code using a Moore Machine architecture.
3. **Functional Simulation:** Creating a testbench to apply stimulus and monitor outputs.
4. **Verification:** Comparing the simulation waveforms against the expected state transitions and output logic.
5. **Synthesis (Optional):** Mapping the RTL to a specific FPGA/ASIC technology (out of scope for this specific RTL file).

3 Functional Description (The RTL)

The design is implemented as a **Moore Finite State Machine (FSM)**. The state of the machine directly determines the registered outputs, ensuring glitch-free operation.

3.1 State Definitions

The FSM consists of five primary states:

- **IDLE:** Starting state, 0 cents accumulated.
- **INP_5:** 5 cents accumulated.

- **INP_10:** 10 cents accumulated.
- **INP_15:** 15 cents accumulated (Water dispensed).
- **INP_20:** 20 cents accumulated (Water dispensed + Change).

3.2 Timeout Mechanism

A 4-bit counter (`CLK_count`) monitors the duration the machine stays in a single state. If the user does not insert a coin within 15 clock cycles (while not in IDLE), the machine automatically returns to IDLE and returns the currently inserted amount as change (`DISP_CHANGE = '1'`).

3.3 Output Logic

The outputs `DISP_WATER` and `DISP_CHANGE` are registered.

- `INP_15` triggers `DISP_WATER`.
- `INP_20` triggers both `DISP_WATER` and `DISP_CHANGE`.

4 The Testbench and Verification Process

The verification is performed using a behavioral VHDL testbench (`tb_vending`). The testbench acts as a wrapper around the Device Under Test (DUT).

4.1 Verification Strategy

The verification process is divided into three critical test scenarios to ensure total coverage of the logic:

Test Case	Input Stimulus	Expected Output
1. Exact Amount	5c then 10c	Water: '1', Change: '0'
2. Overpay	10c then 10c	Water: '1', Change: '1'
3. Inactivity	10c then wait 16 cycles	Water: '0', Change: '1'

Table 1: *Verification Test Plan*

4.2 Working of the Testbench

1. **Clock Generation:** A continuous 100MHz (10ns period) clock is generated using a simple process.
2. **Reset:** The `CLR` signal is asserted for 2 cycles to ensure the FSM starts in a known IDLE state.
3. **Stimulus Application:** The `stimulus` process drives the `xCOIN` input to mimic real-world interactions.
4. **Observation:** The user monitors the `DISP_WATER` and `DISP_CHANGE` signals on a waveform viewer (like ModelSim or Vivado) to confirm timing and logic accuracy.

5 Key Technical Highlights of the Design

5.1 Moore Machine Architecture

The controller is implemented as a **Moore FSM**, where outputs are determined solely by the current state. This ensures that the dispensing signals (`DISP_WATER` and `DISP_CHANGE`) are stable and synchronized with the clock edge, preventing combinational glitches that could lead to false dispensing.

5.2 Input Encoding and Safety

The design utilizes a **One-Hot Encoding** scheme for the coin inputs (`xCOIN`).

- "001": No Coin
- "010": 5 cents
- "100": 10 cents

This methodology minimizes decoding logic and reduces the risk of metastable states when multiple buttons are pressed simultaneously.

5.3 Inactivity Timeout and Refund Logic

A critical safety feature is the `CLK_count` mechanism. If the FSM remains in a non-IDLE state for more than 15 cycles without receiving a new coin, the machine triggers a timeout.

$$\text{If } CLK_count = TIMEOUT \rightarrow next_state = IDLE, disp_c_reg = '1' \quad (1)$$

This ensures that user funds are automatically returned as change if the transaction is abandoned mid-way.

5.4 Registered Outputs

The use of `disp_w_reg` and `disp_c_reg` allows the system to provide a precise, single-cycle pulse for external hardware (like solenoids or motors), ensuring the physical hardware is not active longer than intended.

6 Conclusion

The design successfully implements a robust vending machine controller. Through the use of a registered Moore machine and a comprehensive testbench, the design ensures that timing requirements are met and that the system handles both successful transactions and user inactivity (timeout) correctly.